Question 1. (5 x 1 = 5 points) Circle around the right answer (TRUE or FALSE) for questions below. Ambiguous answers and unclear markings will result in zero points. You do NOT have to justify your answer for the objective questions in Parts (a) - (e) of Question Q1.

(a) Let A, B, and C be events with $P(C) > 0$. If A and B are independent, then A and B may become dependent when conditioning on C.

$A \rightarrow C \leftarrow B$ {theory of d-separation} $\rightarrow$ depends on conditioning

TRUE / FALSE

(b) Suppose P is the transition matrix of an irreducible Markov chain on a finite state space, and let π be the stationary distribution. Then for all $n \geqslant 1$, we have $\pi P^n = \pi$.

TRUE / FALSE

(c) A random time τ is called a **stopping time** if for each $n \geqslant 1$, the event $\{\tau = n\}$ can be determined by observing only $X_1, X_2, \ldots, X_n$. Define a sequence of random times as follows:

- Let $\tau_1 = \inf\{n \geqslant 1 : X_n > 0\}$
- For $k \geqslant 2$, let $\tau_k = \inf\{n > \tau_{k-1} : X_n > X_{\tau_{k-1}}\}$

Then τ_k is a stopping time for all $k \geqslant 1$.

TRUE / FALSE

(d) Let $X_1, X_2, X_3, \ldots$ be a sequence of random variables defined on the same probability space $(\Omega, \mathcal{F}, P)$. Suppose that X_n converges to 0 **in probability**. It is possible that for **every** sample path $\omega \in \Omega$, the sequence of real numbers $X_1(\omega), X_2(\omega), X_3(\omega), \ldots$ fails to converge to 0?

Time average $\neq$ Ensemble average (unless ergodicity mentioned).

TRUE / FALSE

(e) Consider simple 1D random walks on the integers $\mathbb{Z}$, starting at 0.

- An **asymmetric** random walk moves right with probability $p > 1/2$ and left with probability $q = 1 - p < 1/2$.
- A **symmetric** random walk moves right with probability $1/2$ and left with probability $1/2$.

The asymmetric random walk is **transient**, and the symmetric random walk is **null recurrent**.

TRUE / FALSE

Question 2. (1 + 2 + 2 + 2 + 2 + 2= 11 points) A miner is trapped in a mine containing three doors. The first door leads to a tunnel that takes him to safety after two hours of travel. The second door leads to a tunnel that returns him to the mine after three hours of travel. The third door leads to a tunnel that returns him to the mine after five hours. Due to the complex environment in the mine, the miner cannot distinguish between previously chosen doors and is equally likely to choose any door at all times. Let X denote the time until the miner reaches safety. Let N denote the total number of doors selected before the miner reaches safety. Let T_i denote the travel time corresponding to the i th choice, $i \geqslant 1$.

(a) Give an identity that relates X to N and the T_i .

(b) What is $E[N]$?

(c) What is $E[T_N]$?

(d) What is $E\left[\sum_{i=1}^{N} T_i | N = n\right]$?

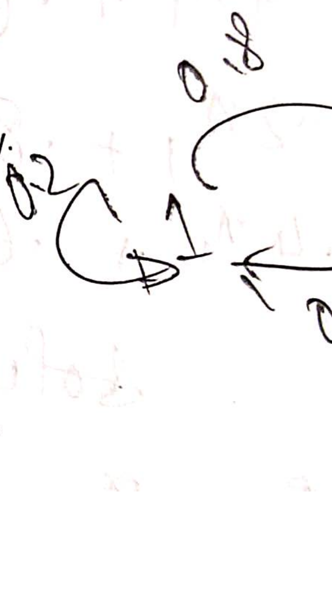
(e) Using the preceding, what is $E[X]$?

(f) Define a Markov chain with states 1 (in the mine), and 2 (safety), and set up the transition probabilities.

Question 3. (2 + 2 + 2 = 6 points) Capa plays either one or two chess games every day, with the number of games that she plays on successive days being a Markov chain with transition probabilities

$$P_{1,1} = 0.2, \quad P_{1,2} = 0.8, \quad P_{2,1} = 0.4, \quad P_{2,2} = 0.6$$

Capa wins each game with probability p. Suppose she plays two games on Monday.



(a) What is the probability that she wins all the games she plays on Tuesday?

(b) What is the expected number of games that she plays on Wednesday?

(c) In the long run, on what proportion of days does Capa win all her games?

Question 4 (2+2+2+2 = 8 points). A small robot explores three rooms: **Room 1**, **Room 2**, and **Room 3**. Every minute, it moves according to the following rules:

- If the robot is in **Room 1**, it always moves to **Room 2**.
- If it is in **Room 2**, with probability $\frac{1}{3}$ it goes back to **Room 1**, and with probability $\frac{2}{3}$ it goes to **Room 3**.
- If it is in **Room 3**, with probability $\frac{1}{2}$ it goes to **Room 2**, and with probability $\frac{1}{2}$ it leaves the system and never returns (we call this "escape").

The robot starts in **Room 1** at time 0. Let X_n be the robot's location after n minutes (with "escape" treated as an absorbing state).

2 (a) Write down the transition matrix for X_n, treating "escape" as a state.

2 (b) What is the expected number of minutes until the robot escapes?

2 (c) What is the probability that the robot ever visits **Room 2** before escaping?

0 (d) What is the probability that the robot visits **Room 3** at least twice before escaping?

Question 5. (1 + 1 points). For a nonnegative integer-valued random variable X, show that:

1 (a) $E[X] = \sum_{n=1}^{\infty} P(X \geqslant n)$.

0.5 (b) Let N denote the first time heads turn up when we repeatedly toss a coin with probability of heads $= p$. Use the answer in Part (a) to determine $E[N]$.

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Question 6. (3 points). A professor continually gives exams to her students. She can give three possible types of exams, and her class is graded as either having done well or badly. Let p_i denote the probability that the class does well on a type i exam, and suppose that $p_1 = 0.3$, $p_2 = 0.6$, and $p_3 = 0.9$. If the class does well on an exam, then the next exam is equally likely to be any of the three types. If the class does badly, then the next exam is always type 1. What proportion of exams are type i, $i = 1, 2, 3$?